

Lesson 3-5 · Day 3 (Observation) · Period F

Multiplicity & Factored-Form Graphing — 5/8/26

Student-centered

Math Objective Construct a factored polynomial from a graph by identifying x -intercepts, choosing the correct multiplicity at each (touch = even, cross = odd), and applying end behavior to determine the leading coefficient sign.

Language Objective Use the frame “*The factor for $x = \underline{\hspace{1cm}}$ must be $\underline{\hspace{1cm}}$ because the graph $\underline{\hspace{1cm}}$* ” to justify each factor of a polynomial.

Essential Understanding A polynomial's graph and its factored form encode the **same information** — zeros locate the factors, multiplicity determines touch vs. cross, and end behavior determines the leading coefficient sign.

DOK 1

5 min

vocabulary recall

Three quick questions — do independently:

1. What is the multiplicity of the zero $x = 4$ in $f(x) = (x - 4)^2(x + 1)$?

2. A zero with **even** multiplicity — the graph _____ the x -axis at that zero. (*touches and turns back? crosses through?*)
3. A zero with **odd** multiplicity — the graph _____ the x -axis at that zero.

DOK 1–2

15 min

I do → you do → I check → we do

Four steps today:

1

I do (4 min) — teacher models with annotation

2

You do (4 min) — independent sketch

3

I check (3 min) — compare with exemplar

4

We do (4 min) — whole-class call-and-response

Launch — I Do (4 min): Watch & Annotate

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DOK 1-2

4 min

teacher models

Teacher example:	$f(x) = (x - 2)^2(x +$
Zero $x = 2$	mult. = ____, behavior: ____
Zero $x = -3$	mult. = ____, behavior: ____
3) Degree	____, leading coeff. sign: ____
End behavior	left ____, right ____

Touch / Cross Rule — copy this on your packet: Even multiplicity → graph **TOUCHES** x-axis and turns back.

Odd multiplicity → graph **CROSSES** through x-axis.

Launch — You Do (4 min): Try This One

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DOK 1–2

4 min

independent work

Sketch $g(x) = (x + 1)(x - 4)^2$ on the grid in your packet.

1. Identify each zero and its multiplicity.
2. Label each zero as **touch** or **cross** using the rule.
3. Determine end behavior and sketch the curve.

No computing exact y -values — sketch the shape only. Use the rule you just copied.

DOK 1–2

3 min

compare with exemplar

Exemplar for $g(x) = (x + 1)(x - 4)^2$

Zero $x = -1$ multiplicity 1 → **CROSS**

Zero $x = 4$ multiplicity 2 → **TOUCH**

Degree 3, pos. coeff. ends: left **down**, right **up**

What did you fix? Write it in your packet. *Common error:* crossing at $x = 4$ instead of touching.

Why it matters: the squared factor $(x - 4)^2$ always bends back — it cannot cross.

DOK 1–2

4 min

whole-class

Class example: $h(x) = -x(x + 2)^2(x - 1)$ — fill in as we

Zeros

$x = \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}$

Touch or cross?

$x = 0$: $\underline{\hspace{1cm}}$ $x = -2$: $\underline{\hspace{1cm}}$ $x = 1$:

go:

$\underline{\hspace{1cm}}$

Degree & sign

degree = $\underline{\hspace{1cm}}$, leading coeff. = $\underline{\hspace{1cm}}$

End behavior

left $\underline{\hspace{1cm}}$, right $\underline{\hspace{1cm}}$

TOUCH / CROSS RULE — the key to today's

EVEN

TOUCHES the x -axis and turns back (*squared, quartic, ...*)

task:

ODD

CROSSES through the x -axis (*linear, cubic, ...*)

SIGN

Leading coeff. $+$ $\rightarrow$ ends **up-right**; leading coeff. $-$ $\rightarrow$ ends **down-right**

Quick self-check before Explore: Read a factor, name its multiplicity, call touch or cross — in 3 seconds.

If you cannot, revisit the rule card in your packet *now*.

Explore — Mystery Graph DOK-3 (10 min TPS)

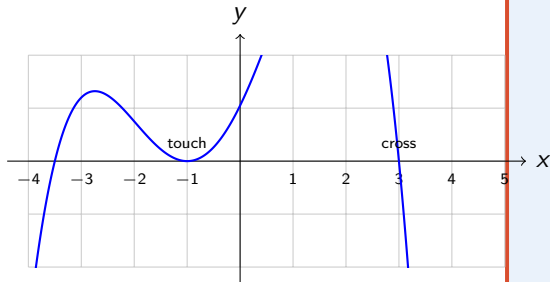
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DOK 3

10 min

think-pair-share

The Graph



Your task: Build a possible factored form.

Justify every piece.

1. Factor for $x = -1$: why?
2. Factor for $x = 3$: why?
3. Leading coefficient sign: why?
4. Could the degree differ?

3 min alone → 3 min partner → 4 min whole-class

Explore — Mystery Graph: Whole-Group Synthesis

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DOK 3

4 min

whole-group debrief

Press on these in the debrief:

1. Where does the graph **touch** vs. **cross**? What does each tell you about the exponent of that factor?
2. What does **both ends down** tell you about the degree and leading coefficient sign?
3. Why must the leading coefficient be **negative**?
4. Could the degree be 4 instead of 3? What would change?
5. How would you check your factored form using Desmos?

Explore — DOK-2 Consolidation (15 min, partners) Algebra 2 · Lesson 3-5

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DOK 2

15 min

partner work

Three items in order — work with your partner:

1. **(Savvas q12)** Sketch $f(x) = 3x^3 - 9x^2 - 12x$. Find zeros first (factor it), then label touch / cross.
2. **(Try-It 2b)** Describe behavior at each zero of $f(x) = (x^2 + 9)(x - 1)^5(x + 2)^2$. What is $(x^2 + 9)$?
3. A graph crosses at $x = 0$, touches at $x = 2$, both ends up. Write a possible factored form. Justify the leading coeff. sign.

Record reasoning, not just answers. Verify with Desmos at the end of each item.

Use these frames throughout today's tasks:

1. "The factor for $x =$ ____ must be ____ because the graph ____."
2. "The leading coefficient is ____ because the ends go ____."
3. "I notice ____, so I think ____."

DOK 1-2

5 min

synthesis

Today's rule, one more time: **Even** multiplicity $\rightarrow$ TOUCH **Odd** multiplicity $\rightarrow$ CROSS

Leading coeff. sign + degree parity $\rightarrow$ END BEHAVIOR

Objectives — which did you hit today?

- ☐ **Math:** Construct a factored polynomial from a graph (zeros + multiplicity + end behavior).
- ☐ **Language:** Used the sentence frame to justify each factor.
- ☐ **Essential Understanding:** Graph and factored form encode the same information.

Exit Ticket — DOK-1 Cool-Down (5 min)

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DOK 1

5 min

not graded — turn in at door

Complete independently on the exit ticket strip:

1. A factor of $(x - 4)^2$ produces a _____ at $x = 4$. *(touch / cross)*
2. A factor of $(x + 1)$ produces a _____ at $x = -1$. *(touch / cross)*
3. **The biggest thing I learned today was:** _____

Thank you for visiting.